

World Trade – An Overview

Krugman, Obstfeld & Melitz, 12th Edition (Chapter 2)

Prepared by: Macro/Trade Lecture Notes

August 28, 2025

Agenda

- 1 Learning Objectives
- 2 Empirical Landscape
- 3 Gravity Model
- 4 Distance and Borders
- 5 Globalization in Two Waves
- 6 Composition of Trade
- 7 Policy and Macro Linkages
- 8 Estimation Notes
- 9 Discussion and Wrap-Up

Learning Objectives

After this lecture, you should be able to:

- Describe how the value of trade between any two countries depends on the size of these countries' economies and explain the reasons for that relationship.
- Describe why distance and borders still matter for trade.
- Summarize the two ages of globalization and their reversals.
- Discuss how the composition of world trade has evolved (goods, services, GVCs).
- Connect empirical patterns to policy and macroeconomic performance.

Stylized Facts of World Trade

- Trade is concentrated among large economies; small and proximate partners trade disproportionately with each other.
- The share of trade in world GDP rose sharply post-1945, with interruptions around major crises.
- Bilateral trade is highly correlated with partners' GDP and negatively related to distance.
- More than 30 percent of world output is sold across national borders.
 - World trade in goods and services exceeded \$25 trillion in 2019.

Stylized Facts of World Trade Cont'd

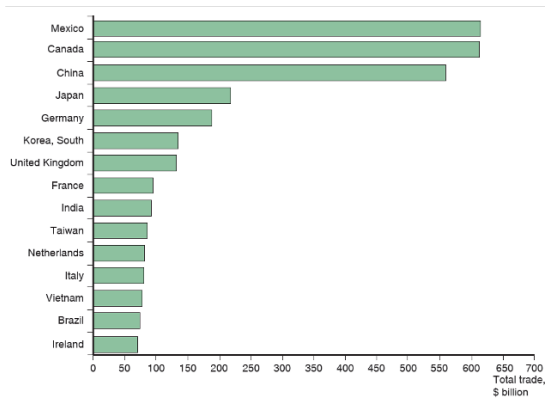


Figure: Total U.S. Trade with Major Partners, 2019

- U.S. trade is measured as the sum of imports and exports - mostly with 15 major partners.

Stylized Facts of World Trade Cont'd

- Figure 2-1 shows the total value of trade in goods exports plus imports between the United States and its top 15 trading partners in 2019.
 - The largest 15 trading partners with the United States accounted for 75 percent of the value of U.S. trade.
- The five largest trading partners with the United States in 2019 were Mexico, Canada, China, Japan, and Germany.

Size Matters, The Gravity Model

- Three of the top 15 U.S. trading partners are European nations: Germany, the United Kingdom, and France.
- Why does the United States trade more with these European countries than with others?
 - These three countries have the largest gross domestic product (G D P), the value of goods and services produced in an economy, in Europe.

A parsimonious empirical relationship:

$$T_{ij} = A \frac{Y_i^\alpha Y_j^\beta}{D_{ij}^\gamma} \times \exp\{\mathbf{z}_{ij}'\delta\},$$

where T_{ij} is trade from i to j , Y_i , Y_j are GDPs, D_{ij} is distance, and $\mathbf{z}_{ij}$ contains frictions (common language, border, trade agreement, etc.).

Log-Linear Specification and Interpretation

Taking logs:

$$\ln T_{ij} = c + \alpha \ln Y_i + \beta \ln Y_j - \gamma \ln D_{ij} + \mathbf{z}_{ij}'\delta + \varepsilon_{ij}.$$

- $\alpha, \beta \approx 1$: trade proportional to size.
- $\gamma > 0$: distance elasticity; often near unity in estimates.
- Elements of $\mathbf{z}_{ij}$: common language (+), border (−), FTA (+), currency union (+).

Structural Gravity (Sketch)

Under CES preferences and iceberg costs $\tau_{ij} \geq 1$:

$$X_{ij} = \frac{Y_i E_j}{Y^W} \left(\frac{\tau_{ij}}{\Pi_i P_j} \right)^{1-\sigma},$$

where X_{ij} is expenditure of j on goods from i , $\sigma > 1$ is elasticity of substitution, Π_i and P_j are multilateral resistance terms, and Y^W world income.

- Captures general equilibrium: bilateral trade depends on *all* trade costs via Π_i, P_j .
- Empirically implemented with exporter and importer fixed effects.

Why Distance Still Matters

- Transportation and logistics: bulky/perishable goods face non-trivial costs.
- Information and trust: proximity reduces uncertainty and search costs.
- Regional production networks: NAFTA/USMCA, EU, ASEAN reinforce neighborhood trade.

Beyond Tariffs: Border Effects

- Legal, regulatory, standards, and customs procedures.
- Currency and financial frictions; exchange rate risk.
- Culture and language; business practices and contract enforcement.

Empirically, border dummies are large in gravity regressions, even holding distance constant.

First Age (pre-1914)

- Technology: steamships, railways, telegraph; sharp drop in trade costs.
- Institutions: classical gold standard enabling stable exchange rates.
- Policy: unilateral tariff reductions (e.g., U.K.) and bilateral treaties.

Retreat and Second Age

- Interwar retreat: wars, Great Depression, protectionism, collapse of gold standard.
- Post-1945 resurgence: Bretton Woods, GATT/WTO, containerization, ICT.
- Trade/GDP exceeded pre-WWI highs; yet reversals occur during crises.

From Commodities to Manufacturing and Services

- Manufacturing now dominates goods trade; services are fastest-growing segment.
- Rise of tradable services: IT, finance, engineering, design, and digital platforms.
- Global value chains (GVCs): fragmentation of production; importance of intermediate inputs.

Implications of GVCs

- Value added is internationally distributed; gross exports overstate domestic content.
- Policy shocks propagate along supply chains; resilience vs. efficiency trade-offs.
- Measurement: input-output methods and value-added trade statistics.

Trade Policy Levers

- Tariffs and non-tariff measures (standards, quotas, rules of origin).
- Infrastructure: ports, customs modernization, digital connectivity.
- Trade agreements and economic unions: depth matters (services, investment, data).

Growth, Development, and Shocks

- Integration into GVCs is associated with faster growth and technology diffusion.
- Weak institutions and infrastructure amplify effective distance.
- Shocks (pandemics, geopolitical events) reveal supply chain fragilities; diversification and friend-shoring emerge as strategies.

- Log-linear OLS on strictly positive flows; Poisson Pseudo-Maximum Likelihood (PPML) handles zeros and heteroskedasticity.
- Fixed effects for exporter/importer/time absorb multilateral resistance and size.
- Robustness: clustering at dyad or country level; instrumenting policy variables when needed.

Interpreting Coefficients

- Distance elasticity ($-\gamma$): percent change in trade for a percent change in distance.
- Border dummy: multiplicative effect on bilateral trade beyond geography.
- FTA dummy: treatment effect capturing average agreement impact.

Discussion Questions

- ① Why does distance retain explanatory power in the digital era?
- ② How do GVCs complicate the meaning of "exports" and "imports"?
- ③ Is globalization cyclical or inevitable? What determines reversals?
- ④ How should policy balance efficiency with resilience in supply chains?
- ⑤ How might digital services alter effective distance in gravity models?

Key Takeaways

- Gravity provides a powerful empirical baseline consistent with theory.
- Trade costs are multi-dimensional; borders matter beyond tariffs.
- Globalization has come in waves; policy and shocks shape its path.
- Composition of trade is shifting toward services and complex GVCs.

Further Reading

- Krugman, Obstfeld, and Melitz (12th ed.), Chapter 2.
- Anderson, J. E. and van Wincoop, E. (2003), "Gravity with Gravitas," *AER*.
- Head, K. and Mayer, T. (2014), "Gravity Equations: Workhorse, Toolkit, and Cookbook," in *Handbook of International Economics*.

Questions?

(End of Lecture)